Función de autocovarianza y autocorrelación del proceso AR(2)

Calculemos la función de autocovarianza y autocorrelación del proceso AR(2)

$$\gamma_X(h) = \mathrm{Cov}(\tilde{X}_t, \tilde{X}_{t+h}) = \mathbb{E}(\tilde{X}_t \tilde{X}_{t+h}) - \mathbb{E}(\tilde{X}_t)\,\mathbb{E}(\tilde{X}_{t+h})$$

Además, sabemos que:

$$\mathbb{E}(\tilde{X}_t) = \mathbb{E}(X_t - \mu) = \mathbb{E}(X_t) - \mu = 0$$

Por lo tanto,

$$\gamma_X(h) = \mathbb{E}(\tilde{X}_t \tilde{X}_{t+h})$$

$$\tilde{X}_{t+h} = \phi_1 \tilde{X}_{t+h-1} + \phi_2 \tilde{X}_{t+h-2} + Z_{t+h}$$

Entonces:

$$\tilde{X}_t \tilde{X}_{t+h} = \phi_1 \tilde{X}_t \tilde{X}_{t+h-1} + \phi_2 \tilde{X}_t \tilde{X}_{t+h-2} + \tilde{X}_t Z_{t+h}$$

> **Recordemos** que el proceso AR(2) considerado es:
>
> $$(1 - \phi_1 B - \phi_2 B^2)\tilde{X}_t = Z_t$$
>
> $$\tilde{X}_t = X_t - \mu$$
>
> $$\tilde{X}_t - \phi_1 \tilde{X}_{t-1} - \phi_2 \tilde{X}_{t-2} = Z_t$$

Ahora evaluamos:

$$\mathbb{E}(\tilde{X}_t \tilde{X}_{t+h}) = \phi_1 \mathbb{E}(\tilde{X}_t \tilde{X}_{t+h-1}) + \phi_2 \mathbb{E}(\tilde{X}_t \tilde{X}_{t+h-2}) + \mathbb{E}(\tilde{X}_t Z_{t+h})$$

Si $h = 0$:

$$\gamma_X(0) = \phi_1 \mathbb{E}(\tilde{X}_t \tilde{X}_{t-1}) + \phi_2 \mathbb{E}(\tilde{X}_t \tilde{X}_{t-2}) + \mathbb{E}(\tilde{X}_t Z_t)$$

$$\mathbb{E}(\tilde{X}_t Z_t) = \phi_1 \mathbb{E}(\tilde{X}_{t-1} Z_t) + \phi_2 \mathbb{E}(\tilde{X}_{t-2} Z_t) + \mathbb{E}(Z_t^2)$$

Como Z_t es ruido blanco e independiente de $\tilde{X}_{t-1}$ y $\tilde{X}_{t-2}$:

$$\mathbb{E}(\tilde{X}_{t-1} Z_t) = 0, \quad \mathbb{E}(\tilde{X}_{t-2} Z_t) = 0, \quad \mathbb{E}(Z_t^2) = \sigma_Z^2$$

Entonces:

$$\mathrm{Var}(Z_t) = \sigma_Z^2 = \mathbb{E}(Z_t^2) - \mathbb{E}(Z_t)^2$$

Por lo tanto:

$$\gamma_X(0) = \phi_1 \gamma_X(1) + \phi_2 \gamma_X(2) + \sigma_Z^2$$

Si $|h| > 0$:

$$\gamma_X(h) = \phi_1 \gamma_X(h-1) + \phi_2 \gamma_X(h-2)$$

Por lo tanto:

$$\gamma_X(h) = \begin{cases} \phi_1 \gamma_X(1) + \phi_2 \gamma_X(2) + \sigma_Z^2 & \text{si } h = 0 \\ \phi_1 \gamma_X(h-1) + \phi_2 \gamma_X(h-2) & \text{si } |h| > 0 \end{cases}$$

Ahora calculemos la FAC teórica del proceso AR(2)

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)}$$

$$\rho_X(1) = \rho_1 = \frac{\gamma_X(1)}{\gamma_X(0)} = \frac{\phi_1 \gamma_X(0) + \phi_2 \gamma_X(1)}{\gamma_X(0)} = \phi_1 + \phi_2 \frac{\gamma_X(1)}{\gamma_X(0)} = \phi_1 + \phi_2 \rho_1$$

Entonces:

$$\rho_1 = \phi_1 + \phi_2 \rho_1 \Rightarrow \rho_1 = \frac{\phi_1}{1 - \phi_2}$$

Por otra parte,

$$\rho_X(2) = \rho_2 = \frac{\gamma_X(2)}{\gamma_X(0)} = \phi_1 \frac{\gamma_X(1)}{\gamma_X(0)} + \phi_2 = \phi_1 \rho_1 + \phi_2$$

Es decir:

$$\rho_2 = \phi_1 \rho_1 + \phi_2$$

En general:

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2}, \quad \text{para } k \geq 3$$

Considere el proceso AR(2) del ejemplo anterior, es decir:

$$(1 - 0.9B + 0.2B^2) X_t = Z_t \quad ; \quad \{Z_t\} \sim \mathrm{wn}(0, \sigma_Z^2)$$

Donde:

$$\phi_1 = 0.9 \quad ; \quad \phi_2 = -0.2$$

Calculemos las 3 primeras autocorrelaciones para este proceso: (ρ_1, ρ_2, ρ_3)

$$\rho_1 = \frac{\phi_1}{1-\phi_2} = \frac{0.9}{1+0.2} = 0.75$$

$$\rho_2 = (0.9)\rho_1 + (-0.2) = 0.675 - 0.2 = 0.475$$

$$\rho_3 = 0.9\rho_2 + (-0.2)\rho_1 = 0.9(0.475) - 0.2(0.75) = 0.2775$$

Gráfica de la FAC teórica del proceso AR(2) *(1 - 0.9B + 0.2B^2)X_t = Z_t*

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'https://nuriaarroyo.github.io/ecoII/interactive/ACF_te%C3%B3rica_AR_1_1_-_0_4B_X_t_Z_t_%CF%8
6_0_4.html'
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